AIIDE 2026 POSTER SESSION

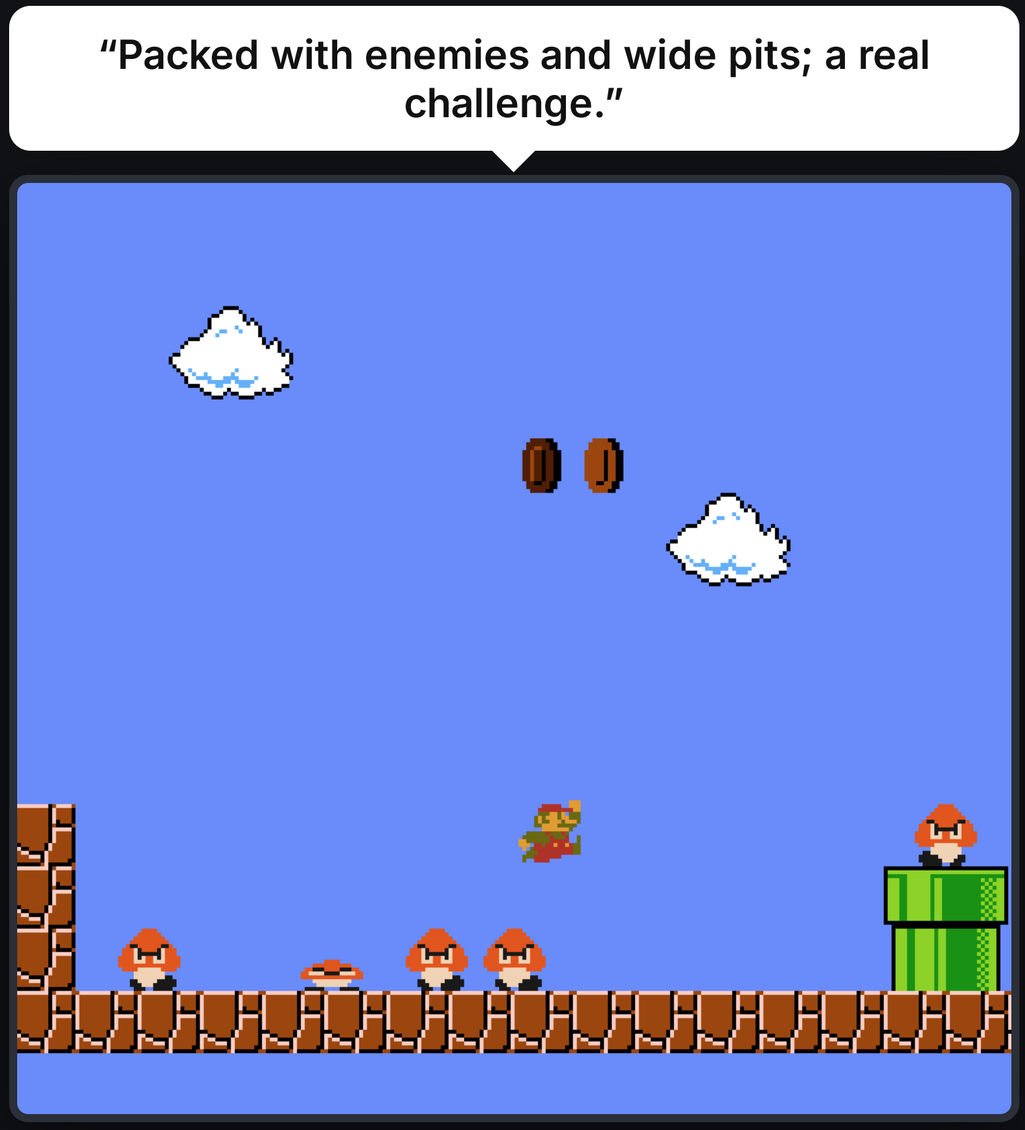
Did You Get the Level You Asked For?

Bridging LLMs and PCGRL for *Super Mario Bros.* Level Generation

Bruno Lemos · Luiz Chaimowicz

Universidade Federal de Minas Gerais (UFMG), Brazil

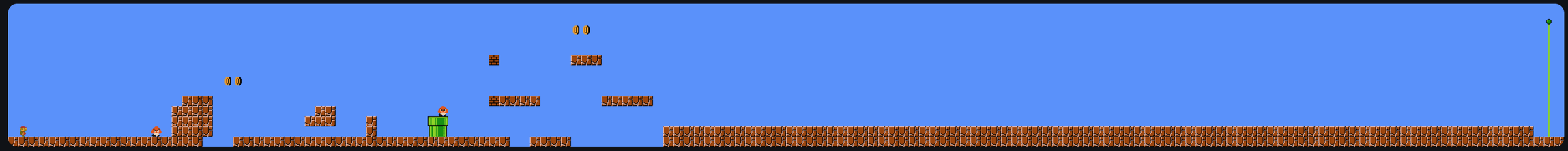
UFMG DCC VERLab Computer Vision and Robotics



generated from the request above · played by an A* agent

You ask for a flat, empty, smooth level…

Corpus generator | Ours



`error 0.16` enemies, gaps, coins and hills you did not ask for

> **Generators are material-bound.** If every corpus chunk has an enemy, no recombination of chunks has none.

Separate what was asked from how it is built

A

Language bridge

GPT-5.2 turns the request into **8 numbers**, t ∈ [0,1]⁸.



B

Compose

A **MaskablePPO** policy stitches ~10 real corpus chunks.



C

Gated edit

The **same policy** applies macro-edits. Any edit that breaks the level is **reverted**.



D

Polish + A* gate

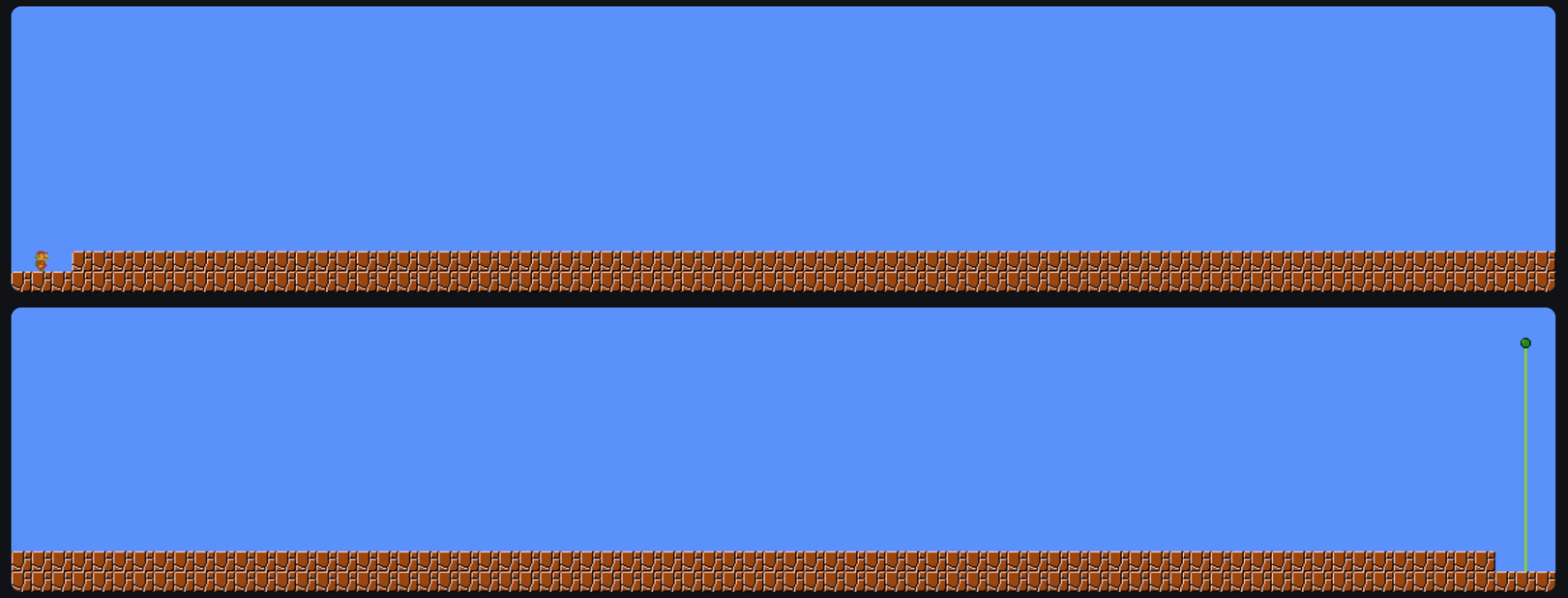
Exact metric fixes, then an **A* agent** must finish the level.



The LLM **never touches a tile**. So we can measure translation error and generation error **separately**.

Watch one episode: compose, then edit

flat & empty | hilly & rich | max chaos



POLISH

mean intent error **0.000**

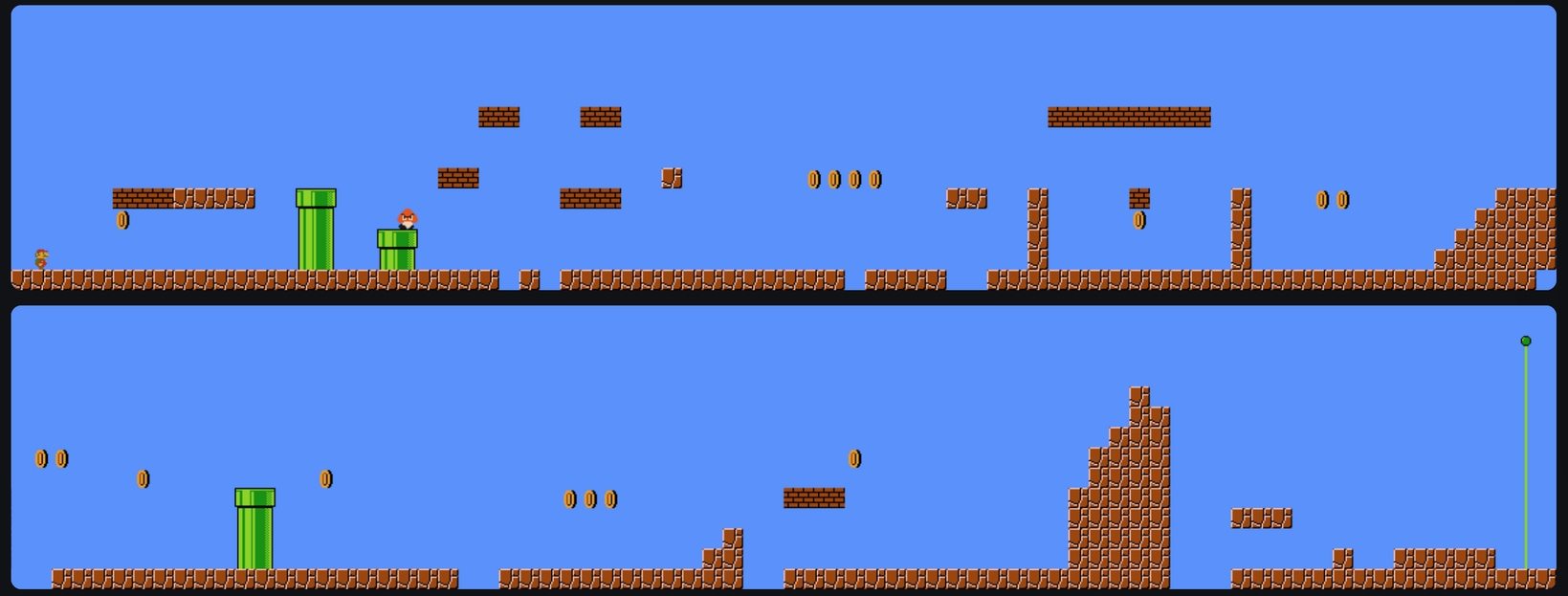
enemies	0.00	
coins	0.00	
blocks	0.00	
pipes	0.00	
gaps	0.00	
verticality	0.00	
smoothness	1.00	
platforms	0.00	

target | measured

Ask, and get (recorded GPT-5.2 translations, final 10M-step policy)

Coin heaven | Vertical climb | Pipe garden | Real challenge | Peaceful stroll | Max chaos | Hilly & rich | Speedrunner's dream

“Coin heaven with barely any danger.”



mean intent error 0.022

✓ completed by the A* agent



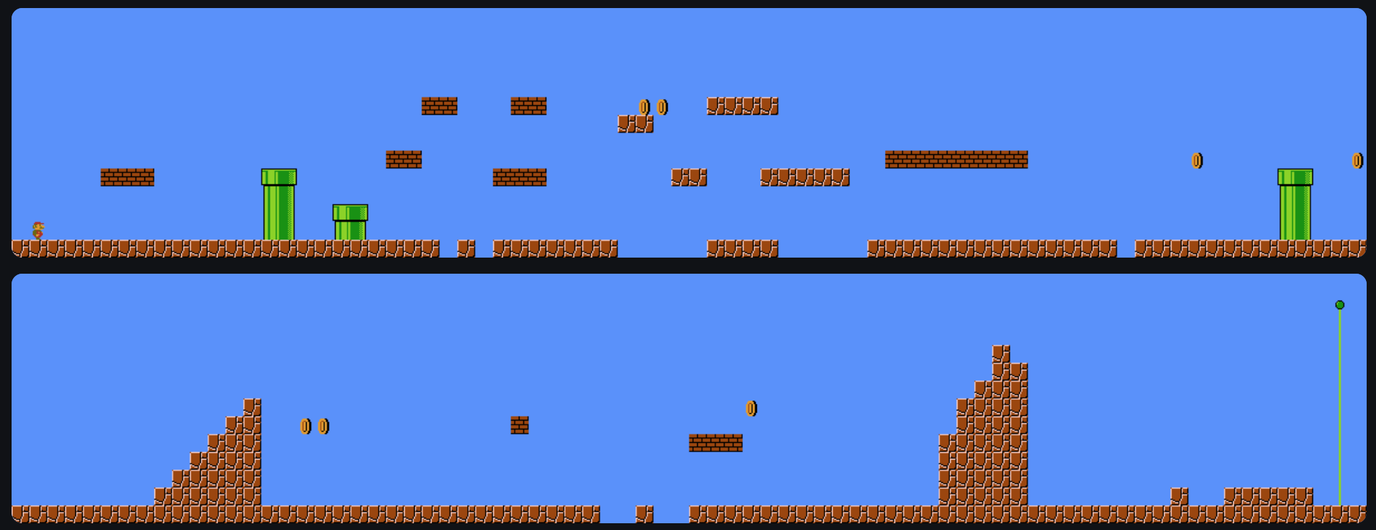


Control reaches the boundaries, including zero

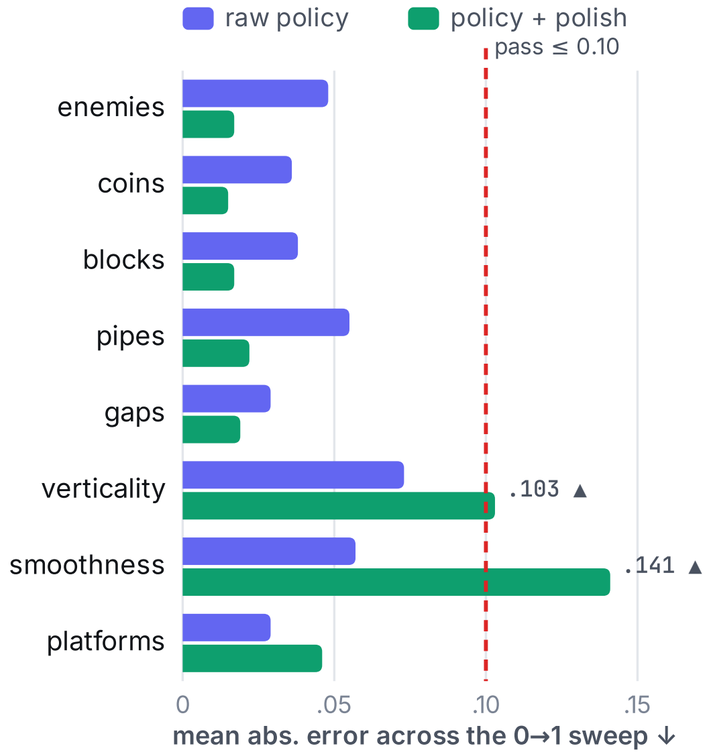
enemies | coins | gaps | verticality

target `0.00` → measured `0.00`

0 .25 .5 .75 1



8/8 dimension sweeps pass
(MAE ≤ 0.10, Spearman ρ ≥ 0.8)

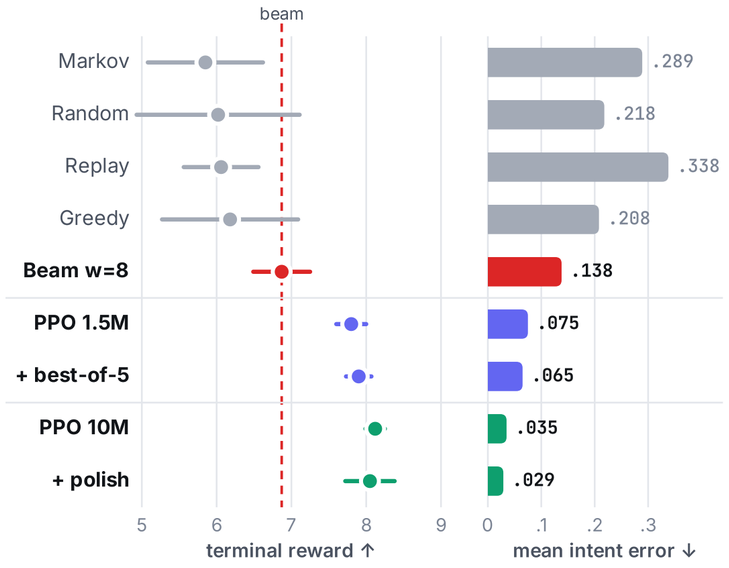


Results in four numbers

RQ1 · LEARNING VS SEARCH

+1.0

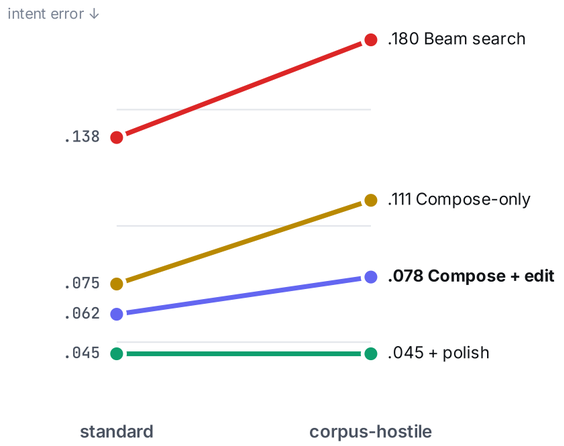
reward over beam search on all 3 seeds ($p = 10^{-4}$), **7× faster** (64 ms vs 0.47 s)



RQ3 · OFF-CORPUS REQUESTS

−30%

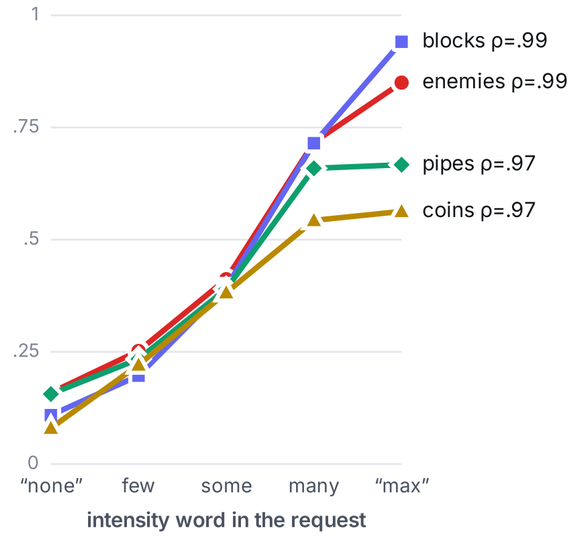
error vs compose-only on corpus-hostile targets; polish reaches **0.004** on flat-empty



RQ4 · LANGUAGE

0.053

translation MAE over 46 descriptions, monotonic end-to-end (ρ = 0.97–0.99)



The request becomes a numeric contract.

- **Translation** missed? Inspect the vector.
- **Generator** missed? Inspect per-dimension residuals.
- **Targets conflict?** Reported, not hidden.

Scope: structural control over a fixed 8-dim vocabulary · one game · no designer study yet.

Slides, demo & code

`github.com/brunolemosdl/intent-pcgrl`

Come talk to us at the poster!